

Decomposing Modal Collapse: A Four-Valued Analysis of Gödel–Scott Positivity

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Abstract

I study the inferential architecture of the Gödel–Scott ontological theory under a bilateral four-valued semantics admitting both inconsistent and incomplete information. Scott’s classical theory is fixed as a control system and reconstructed incrementally over a Belnap–Dunn/FDE propositional kernel, bilateral Kripke modality, actualist individual quantification, and higher-order property quantification. The original conjecture that positive and negative modal collapse might separate is falsified under involutive FDE negation: the informative signed collapse schemata are equivalent by substitution. The substantive decomposition instead occurs upstream. Scott’s A1 splits into directional information flows with distinct roles, mere positive support is insufficient for the T1 analogue, and two explicit S5 models show that natural bilateral essence does not make support-based Godlikeness an essence automatically: positive T2 fails through separate glut and gap mechanisms. A relevant consistency/completeness package recovers T2, and Lean 4.30.0 machine-checks the implication. Gate 8 then compares alternative architectures. Once Scott T2 is available, symmetry alone suffices for positive T3, while a non-symmetric S4 model refutes the corresponding replacement. Project-internal exact positive Godlikeness internalizes local reflection, and a literature-grounded bilateral Anderson candidate admits positive necessary Godlike existence without positive modal collapse. The Fitting comparison reveals a further structural boundary. A naive unrestricted bilateral extension domain makes the relevant Fitting consistency package incompatible with any positive Godlike witness; Lean proves this comprehension obstruction directly. I therefore reconstruct the substantive Fitting candidate over a selected FDE-negation-closed domain of admissible extensions, without globally excluding gluts. In that non-vacuous setting, the extensional Godlikeness-as-essence theorem requires no positivity-rigidity premise, the de-re possibility-to-necessity route requires no frame condition, and de-dicto lifting requires only stability of the positive Godlikeness extension rather than full bilateral extension stability. The classification-based Fitting recovery package can be weakened so that exemplification consistency is required only on the negatively classified branch of the A1-L proof; restoring A1-R reconstructs the stronger consistency condition.

A second recovery route bypasses positivity completeness entirely. I derive its Godlike-indiscernibility premise from a quotient-style saturation principle on positive-property profiles. Negation closure upgrades this to bilateral factorization, and quotient-respecting extensions are closed under the basic FDE operations, while a finite countermodel shows that ordinary FDE algebra closure alone does not generate the quotient. The same failure persists after adding A2-style positivity monotonicity and even full domain-level entailment closure. Indeed, Lean shows that naively closing an admissible domain containing FDE bottom under arbitrary global entailment forces unrestricted bilateral comprehension and recreates the Godlike-witness obstruction under relevant consistency. The quotient admits a more canonical formulation instead: profile saturation is the fixed-point condition of an extensive, monotone, idempotent closure operator giving the least quotient-respecting extension above any bilateral extension. Actualist extensional entailment commutes with this closure when actual existence respects the same profile classes, and a finite countermodel shows that the entailment descent can fail when existence splits a profile class. By contrast, an ultrafilter-style complement-decision principle with relevant consistency or A1-R reconstructs the earlier classification route. Finite witnesses also separate positive-only extension stability from its stronger bilateral predecessor. A new rigidity bridge shows that forward admissible positivity transport derives backward positive Godlikeness reflection, while converse positivity transport along existing edges derives forward persistence without requiring frame symmetry. Symmetry plus forward rigidity entails this converse transport, but 32 asymmetric finite models satisfy both transport directions, showing that accessibility symmetry is stronger than the de-dicto proof needs. A complete-S5 Fitting model has positive necessary Godlike existence, genuine inconsistent extension information, and failure of positive modal collapse. The remaining questions concern a principled actualist quotient condition, four-valued filter structure on the closure fixed points, further G-triggered restriction of converse positivity transport, and generalization beyond relational control semantics.

1 Introduction

Gödel’s ontological argument and Dana Scott’s well-known modification form an unusually compact laboratory for higher-order modal reasoning. Their interest for the present paper is structural rather than theological. The theory combines a primitive predicate of positive properties, modal rigidity, higher-order quantification, essence, and necessary existence, while its derivational behavior has been studied extensively with automated and interactive theorem provers [4, 3]. In Scott’s variant the main existence theorem is accompanied by the well-known phenomenon of *modal collapse*, schematically

$$\chi \rightarrow \Box\chi,$$

according to which every truth is necessary.

Recent machine-checked work sharpens the diagnosis of this phenomenon. In particular, the existence of an ultrafilter-like structure among positive properties does not by itself force collapse; rigidity of positivity is a principal modal ingredient [2]. This raises a natural question that classical two-valued semantics cannot express: which information flows inside positivity, essence, and necessary existence are actually responsible for reconstructing the collapse mechanism?

I study this question in a bilateral four-valued setting. Instead of assigning a single Boolean value to a formula, I track positive and negative support independently. A formula may therefore be supported only, opposed only, both supported and opposed, or neither. This gives the four values

$$\mathbf{T} = (1, 0), \quad \mathbf{F} = (0, 1), \quad \mathbf{B} = (1, 1), \quad \mathbf{N} = (0, 0),$$

following the Belnap–Dunn/FDE tradition [1, 5].

The central research question is:

Research Question 1.1 (Upstream structure of modal collapse). Under a four-valued bilateral modal semantics, which components of positivity–negation transfer, positivity rigidity, informational regularity, essence, and higher-order witness structure reconstruct the classical Gödel–Scott modal-collapse mechanism?

The informative signed collapse pair is

$$\text{MC}^+ : \quad +\chi \Rightarrow +\Box\chi, \quad \text{MC}^- : \quad -\chi \Rightarrow +\Box\neg\chi.$$

These are not independent in the fixed FDE setting. Since $-\chi$ is equivalent to $+\neg\chi$, MC^- is exactly MC^+ instantiated at $\neg\chi$, and conversely by involutive negation. The original collapse-separation conjecture is therefore falsified rather than preserved by an ad hoc semantic modification.

The decomposition instead occurs upstream. Scott’s A1 splits into two independent directional information flows,

$$A1_L : \quad -P(\varphi) \Rightarrow +P(\neg\varphi),$$

$$A1_R : \quad +P(\neg\varphi) \Rightarrow -P(\varphi).$$

The two directions acquire different jobs. $A1_R$, together with a semantic positive-support form of A2, yields possible exemplification for truth-only positive properties. $A1_L$ participates in the reflection and essence branch. A positivity glut blocks the unrestricted T1 analogue because the negative support produced by the classical reductio may already be present.

A natural bilateral reconstruction of Scott’s essence definition does not make Godlikeness an essence automatically. Two finite S5 models satisfy the

earlier control axioms, strong A1, and positive rigidity while refuting positive T2: one fails through inconsistent exemplification and one through incomplete positivity information. A targeted regularity package recovers positive T2, after which the necessary-existence branch becomes comparatively robust. Positive support for A5 suffices for a positive T3 in S5, even if necessary existence itself is glutty.

Once positive T2 and T3 are available, modal collapse follows directly from essence and constant-property embedding. The central non-classical difficulty is therefore no longer modal collapse itself, nor even T3. It is the weakest principled recovery of Godlikeness-as-essence from the lower-level four-valued Scott machinery.

Gate 7 machine-checks this dependency picture. Lean 4.30.0 verifies the collapse-channel equivalence, the truth-only T1 theorem, the sufficient T2 recovery theorem, conditional positive T3, witness supply, the essence-compressed positive-collapse theorem, and bilateral classical-recovery interfaces. An executable finite-model layer retains the glut/gap countermodels and establishes bounded individual indispensability of the four current T2 recovery assumptions in a complete generated two-world, one-entity family. The remaining T2-minimality question is therefore explicitly global rather than a claim inferred from bounded search.

The contribution sought is not a “stronger proof of God”. The intended contribution is an explanatory dependency analysis showing where four-valued inconsistency and incompleteness alter the inferential architecture of the Scott theory, including negative results and countermodels that locate hidden classical regularity assumptions.

Methodological discipline. The development proceeds conservatively. I freeze a classical Scott baseline, fix the propositional and modal semantics, and then lift individual Gödel–Scott principles only when their semantic prerequisites are explicit. Failed conjectures are retained as results rather than repaired by changing the semantics after the fact. Implication-sensitive clauses are represented by signed semantic entailment relations rather than by prematurely choosing a four-valued object-language conditional.

Status of this manuscript. Version 0.7 reports Gates 0–7. In addition to the semantic reconstruction through T3 and modal collapse, it records the Lean proofs, executable countermodels, bounded T2 assumption search, and interface-level classical recovery. The next phase compares semantic variants, beginning with support-based versus exact positive Godlikeness, before moving to literature-motivated variants and weaker modal frames.

2 Classical Scott Baseline

STATUS: ESTABLISHED AT THE SEMANTIC LEVEL.

I use Scott’s variant as the classical control theory, following the modern formal presentation of Benzmüller and Scott [4]. The reference environment uses intensional properties over possible worlds, actualist quantification over individuals, possibilist quantification over properties, and an S5 reference frame. Later experiments may weaken the modal frame, but such changes are not part of the baseline.

Let $P(\varphi)$ mean that property φ is positive. The relevant axioms and definitions are:

$$\text{A1} \quad \neg P(\varphi) \leftrightarrow P(\neg\varphi), \quad (1)$$

$$\text{A2} \quad P(\varphi) \wedge \Box \forall^E y (\varphi(y) \rightarrow \psi(y)) \rightarrow P(\psi), \quad (2)$$

$$\text{D1} \quad G(x) \equiv \forall \varphi (P(\varphi) \rightarrow \varphi(x)), \quad (3)$$

$$\text{A3} \quad P(G), \quad (4)$$

$$\text{A4} \quad P(\varphi) \rightarrow \Box P(\varphi), \quad (5)$$

$$\text{D2} \quad \varphi \text{ Ess } x \equiv \varphi(x) \wedge \forall \psi (\psi(x) \rightarrow \Box \forall^E y (\varphi(y) \rightarrow \psi(y))), \quad (6)$$

$$\text{D3} \quad NE(x) \equiv \forall \varphi (\varphi \text{ Ess } x \rightarrow \Box \exists^E y \varphi(y)), \quad (7)$$

$$\text{A5} \quad P(NE). \quad (8)$$

The standard theorem chain includes

$$\text{T1} \quad P(\varphi) \rightarrow \Diamond \exists^E x \varphi(x), \quad (9)$$

$$\Diamond \exists^E x G(x), \quad (10)$$

$$\text{T2} \quad G(x) \rightarrow G \text{ Ess } x, \quad (11)$$

$$\text{T3} \quad \Box \exists^E x G(x), \quad (12)$$

$$\text{MC} \quad \chi \rightarrow \Box \chi. \quad (13)$$

For the present project, T3 is a control theorem while MC is the main structural target. A proposed four-valued theory must state precisely in what sense this classical system is recovered when all relevant valuations are restricted to the consistent and complete values **T** and **F**.

The project-level baseline, including quantifier conventions and dependency notes, is maintained in `docs/SCOTT_BASELINE.md`.

3 Four-Valued Propositional Kernel

STATUS: ESTABLISHED AT THE SEMANTIC LEVEL.

The propositional base follows the information structure of Belnap–Dunn First-Degree Entailment [1, 5]. Every formula φ receives two independent

bits

$$v(\varphi) = (t(\varphi), f(\varphi)) \in \{0, 1\}^2,$$

where $t(\varphi) = 1$ represents positive support and $f(\varphi) = 1$ negative support. I write $+\varphi$ and $-\varphi$ for these two satisfaction conditions.

Value	Pair	Reading
T	(1, 0)	positive support only
F	(0, 1)	negative support only
B	(1, 1)	both positive and negative support
N	(0, 0)	neither positive nor negative support

Atoms receive positive and negative support independently. The connectives are fixed by the bilateral clauses

$$\begin{aligned}
+\neg\varphi &\iff -\varphi, & -\neg\varphi &\iff +\varphi, \\
+(\varphi \wedge \psi) &\iff +\varphi \wedge +\psi, & -(\varphi \wedge \psi) &\iff -\varphi \vee -\psi, \\
+(\varphi \vee \psi) &\iff +\varphi \vee +\psi, & -(\varphi \vee \psi) &\iff -\varphi \wedge -\psi.
\end{aligned}$$

Positive semantic consequence preserves positive satisfaction:

$$\Gamma \models_{\text{FDE}} \varphi$$

iff every valuation positively satisfying every member of Γ also positively satisfies φ . Equivalently, the positively designated values are $\{\mathbf{T}, \mathbf{B}\}$. Negative satisfaction remains separately observable and will be used for bilateral meta-conditions in later sections.

Crucially, no object-language implication is fixed at this stage. Although $\neg\varphi \vee \psi$ is definable, treating it as *the* conditional would prematurely determine the lifting of A2, D1, D2, and D3.

Proposition 3.1 (Paraconsistency). *The kernel is non-explosive: $p, \neg p \not\models_{\text{FDE}} q$.*

Proof. Let $v(p) = \mathbf{B}$ and $v(q) = \mathbf{N}$. Then p and $\neg p$ are both positively satisfied, while q is not. $\square$

Proposition 3.2 (Paracompleteness). *Excluded middle is not valid: $\not\models_{\text{FDE}} p \vee \neg p$.*

Proof. Let $v(p) = \mathbf{N}$. Negation preserves $\mathbf{N}$, so neither disjunct is positively satisfied and hence neither is their disjunction. $\square$

Proposition 3.3 (Classical recovery of the propositional fragment). *If atomic valuations are restricted to $\{\mathbf{T}, \mathbf{F}\}$, then the clauses for $\neg$, $\wedge$, and $\vee$ remain inside $\{\mathbf{T}, \mathbf{F}\}$ and coincide with the classical truth tables.*

Proof. Immediate by structural induction on formulas using the bilateral clauses above. $\square$

The full gate specification is maintained in `docs/FOUR_VALUED_KERNEL.md`.

4 Four-Valued Modal Semantics

STATUS: ESTABLISHED AT THE SEMANTIC LEVEL.

The modal layer is fixed in two stages. For the first Gödel–Scott decomposition experiments I use a conservative bilateral Kripke semantics as a control. I then represent this semantics inside a paired-neighborhood framework that can later be generalized without changing the four-valued propositional kernel. This separates two possible sources of non-classical behavior: four-valued information and non-Kripke modal structure.

4.1 Relational control semantics

Let $\mathcal{F} = (W, R)$ be a Kripke frame. Positive and negative satisfaction for necessity are defined by

$$\begin{aligned} w \models^+ \Box \varphi &\iff \forall v (wRv \Rightarrow v \models^+ \varphi), \\ w \models^- \Box \varphi &\iff \exists v (wRv \wedge v \models^- \varphi). \end{aligned}$$

Possibility uses the converse quantifier pattern:

$$\begin{aligned} w \models^+ \Diamond \varphi &\iff \exists v (wRv \wedge v \models^+ \varphi), \\ w \models^- \Diamond \varphi &\iff \forall v (wRv \Rightarrow v \models^- \varphi). \end{aligned}$$

These clauses belong to the established family of Belnap–Dunn modal semantics [6, 8]; recent work also develops an S5-based four-valued setting for essence and accident modalities [7].

Proposition 4.1 (Bilateral modal duality). *Under the relational clauses,*

$$\Diamond \varphi \equiv \neg \Box \neg \varphi \quad \text{and} \quad \Box \varphi \equiv \neg \Diamond \neg \varphi$$

hold at both positive and negative satisfaction levels.

Proof. For example,

$$\begin{aligned} w \models^+ \neg \Box \neg \varphi &\iff w \models^- \Box \neg \varphi \\ &\iff \exists v (wRv \wedge v \models^- \neg \varphi) \\ &\iff \exists v (wRv \wedge v \models^+ \varphi) \\ &\iff w \models^+ \Diamond \varphi. \end{aligned}$$

The remaining signed equivalences are analogous. $\square$

Proposition 4.2 (Classical recovery of the relational modal fragment). *If atomic valuations are restricted to $\{\mathbf{T}, \mathbf{F}\}$, then formulas generated with $\neg, \wedge, \vee, \Box, \Diamond$ remain in $\{\mathbf{T}, \mathbf{F}\}$, and the signed modal clauses coincide with ordinary Kripke semantics.*

Proof. The propositional step is established in Section 3. For a classically valued formula φ , positive and negative satisfaction are complementary at every world. Hence $\Box\varphi$ is positively satisfied exactly when every accessible world positively satisfies φ , and negatively satisfied exactly when some accessible world negatively satisfies it. Exactly one of these conditions holds. The diamond case is dual. $\square$

If R is an equivalence relation, the classical fragment therefore recovers the S5 environment used as the Scott control setting.

4.2 Modal aggregation of gluts and gaps

The signed quantifier pattern retains the full four-valued range. For $\Box\varphi$, value $\mathbf{B}$ occurs when every accessible world positively supports φ while at least one also negatively supports it; value $\mathbf{N}$ occurs when positive support fails at some accessible world while negative support occurs nowhere. Dually, $\Diamond\varphi = \mathbf{B}$ when positive support occurs somewhere and negative support occurs everywhere, while $\Diamond\varphi = \mathbf{N}$ when positive support occurs nowhere and negative support fails somewhere.

Thus modalization does not erase the distinction between inconsistency and incompleteness.

4.3 Paired-neighborhood representation

For a formula φ , write

$$\llbracket\varphi\rrbracket^+ = \{w \in W \mid w \models^+ \varphi\}, \quad \llbracket\varphi\rrbracket^- = \{w \in W \mid w \models^- \varphi\}.$$

At each world introduce two families $\mathcal{U}_w, \mathcal{H}_w \subseteq \mathcal{P}(W)$, interpreted provisionally by their modal roles: $\mathcal{U}$ is a universal/core family and $\mathcal{H}$ a hit/witness family. Define

$$\begin{aligned} w \models^+ \Box\varphi &\iff \llbracket\varphi\rrbracket^+ \in \mathcal{U}_w, & w \models^- \Box\varphi &\iff \llbracket\varphi\rrbracket^- \in \mathcal{H}_w, \\ w \models^+ \Diamond\varphi &\iff \llbracket\varphi\rrbracket^+ \in \mathcal{H}_w, & w \models^- \Diamond\varphi &\iff \llbracket\varphi\rrbracket^- \in \mathcal{U}_w. \end{aligned}$$

The crosswise use of the two families preserves the same bilateral box/diamond duality for arbitrary $\mathcal{U}$ and $\mathcal{H}$.

Every relational frame canonically induces such a pair by

$$\begin{aligned} \mathcal{U}_w^R &:= \{X \subseteq W \mid R[w] \subseteq X\}, \\ \mathcal{H}_w^R &:= \{X \subseteq W \mid R[w] \cap X \neq \emptyset\}. \end{aligned}$$

Proposition 4.3 (Relational representation). *The paired-neighborhood semantics induced by $\mathcal{U}^R, \mathcal{H}^R$ agrees exactly with the relational bilateral semantics at both signed satisfaction levels.*

Proof. Immediate from the definitions: membership in $\mathcal{U}_w^R$ is exactly universal coverage of $R[w]$, while membership in $\mathcal{H}_w^R$ is exactly nonempty intersection with $R[w]$. $\square$

This establishes a hierarchy rather than a competition between Kripke and neighborhood semantics. The relational control model is a principal paired-neighborhood model.

4.4 Classical recovery beyond relational frames

For arbitrary paired neighborhoods, modalization may itself turn classically valued input into **B** or **N**. To preserve classical bivalence impose the complement-duality condition

$$X \in \mathcal{U}_w \iff W \setminus X \notin \mathcal{H}_w.$$

If φ is classical-valued, then

$$\llbracket \varphi \rrbracket^- = W \setminus \llbracket \varphi \rrbracket^+,$$

so the condition guarantees that exactly one signed clause for $\Box\varphi$ holds, and likewise for $\Diamond\varphi$. Every relationally induced pair satisfies this condition.

Experimental decision. For Gates 3–4 I keep the S5 relational control semantics fixed. This makes any later separation of MC^+ and MC^- attributable to the bilateral lifting of Gödel–Scott positivity rather than to a simultaneous weakening of modal structure. Non-principal and non-complement-dual neighborhood models will be introduced only as a second experimental axis after the local rigidity/collapse map has been established.

The full Gate-2 specification is maintained in `docs/MODAL_LIFT.md`.

5 Bilateral Positivity

STATUS: ESTABLISHED AT THE SEMANTIC LEVEL.

The positivity predicate P is the central non-logical primitive of the Scott theory. In the four-valued reconstruction, statements of the form $P(\varphi)$ carry positive and negative information independently.

5.1 Directional decomposition of A1

Scott's A1 is

$$\neg P(\varphi) \leftrightarrow P(\neg\varphi).$$

A split into two biconditional channel schemas would be redundant: because property negation is involutive and A1 is a schema over all properties, substituting $\neg\varphi$ for φ converts either biconditional into the other. I therefore decompose A1 by direction:

$$A1_L : \quad \neg P(\varphi) \Rightarrow +P(\neg\varphi),$$

$$A1_R : \quad +P(\neg\varphi) \Rightarrow \neg P(\varphi).$$

Here and below $\Rightarrow$ remains a metalanguage implication.

Proposition 5.1 (Directional independence of A1). *Neither $A1_L$ nor $A1_R$ semantically entails the other in the four-valued positivity semantics.*

Proof. For $A1_L$ without $A1_R$, let $v(P(\varphi)) = \mathbf{T}$ and $v(P(\neg\varphi)) = \mathbf{N}$. The antecedents of all $A1_L$ instances on this property pair lack negative support, while the mirrored $A1_R$ instance from $+P(\varphi)$ to $\neg P(\neg\varphi)$ fails. Conversely, let $v(P(\varphi)) = \mathbf{F}$ and $v(P(\neg\varphi)) = \mathbf{N}$. Then the $A1_R$ antecedents lack positive support, whereas $\neg P(\varphi)$ holds without $+P(\neg\varphi)$, so $A1_L$ fails. $\square$

When both directions hold, strong bilateral A1 yields

$$+P(\neg\varphi) \Leftrightarrow \neg P(\varphi)$$

and, by substitution,

$$\neg P(\neg\varphi) \Leftrightarrow +P(\varphi).$$

Thus

$$v(P(\neg\varphi)) = \text{swap}(v(P(\varphi))),$$

so property negation exchanges $\mathbf{T}$ and $\mathbf{F}$ while preserving $\mathbf{B}$ and $\mathbf{N}$.

5.2 A4 and informative negative rigidity

Scott's A4,

$$P(\varphi) \rightarrow \Box P(\varphi),$$

has the direct positive-support lifting

$$R^+ : \quad +P(\varphi) \Rightarrow +\Box P(\varphi).$$

A previously considered negative clause,

$$\neg P(\varphi) \Rightarrow \neg\Box P(\varphi),$$

is not informative on the fixed S5 control frames. By the bilateral modal clause, $\neg\Box\psi$ merely requires some accessible world to negatively support ψ ; reflexivity makes the current world such a witness whenever $\neg\psi$ already holds.

The informative negative-persistence condition is instead

$$R_{\text{nec}}^- : \quad \neg P(\varphi) \Rightarrow +\Box\neg P(\varphi),$$

equivalently

$$\neg P(\varphi) \Rightarrow -\Diamond P(\varphi).$$

Proposition 5.2 (A1 couples the rigidity channels). *Globally assuming $A1_L$, $A1_R$, and R^+ entails R_{nec}^- .*

Proof. Suppose $\neg P(\varphi)$ at a world w . By $A1_L$, $+P(\neg\varphi)$. Applying R^+ to $\neg\varphi$ gives $+\Box P(\neg\varphi)$, so every accessible world positively supports $P(\neg\varphi)$. Global $A1_R$ therefore makes every accessible world negatively support $P(\varphi)$. By the bilateral box clause this is exactly $+\Box\neg P(\varphi)$. $\square$

This proposition is the first direct evidence that A1 is not merely a static complement axiom. In the presence of A4 it transfers modal persistence from the positive information channel into the negative one.

5.3 A local obstruction to classical positivity reflection

For the classical collapse spine it is useful to isolate the minimal support interface

$$D1^+ : \quad +G(x) \wedge +P(\varphi) \Rightarrow +\varphi(x).$$

This is not yet the final four-valued definition of Godlikeness.

The classical proof uses a local reflection move

$$G(x), Z(x) \Rightarrow P(Z).$$

Its signed analogue would be

$$REF^+ : \quad +G(x) \wedge +Z(x) \Rightarrow +P(Z).$$

Even strong A1 together with $D1^+$ does not entail REF^+ .

A glut counterassignment takes

$$v(Z(x)) = \mathbf{B}, \quad v(P(Z)) = \mathbf{F}, \quad v(P(\neg Z)) = \mathbf{T}.$$

Strong A1 is respected, and $D1^+$ requires positive support for $\neg Z(x)$, equivalently negative support for $Z(x)$, which the glut already supplies. Yet $+P(Z)$ fails.

A distinct gap counterassignment takes

$$v(Z(x)) = \mathbf{T}, \quad v(P(Z)) = v(P(\neg Z)) = \mathbf{N}.$$

Strong A1 preserves the positivity gap, $D1^+$ is silent, and again $+P(Z)$ fails.

These are local signed counterassignments to the proof interface, not yet full higher-order countermodels. They expose two different classical assumptions hidden in the ordinary reductio: consistency of the relevant exemplification and completeness of the relevant positivity information. Both dimensions will therefore be tracked explicitly in the collapse analysis.

6 From Positivity Rigidity to Modal Collapse

The classical proof structure identifies a narrow pressure point for the four-valued analysis. Fix a world w , an individual x , and a property Z . In the Scott theory,

$$G(x)@w, Z(x)@w$$

forces $P(Z)@w$: if Z were non-positive, A1 would make $\neg Z$ positive, while D1 would then force the God-like x to exemplify $\neg Z$, contradicting $Z(x)$. A4 then yields $\Box P(Z)$, and D1 propagates Z to God-like entities at accessible worlds. Thus the local classical spine is

$$G(x), Z(x) \xrightarrow{A1+D1} P(Z) \xrightarrow{A4} \Box P(Z) \xrightarrow{D1} \Box \forall z (G(z) \rightarrow Z(z)).$$

Substituting the constant-in-individuals property $Z := \lambda z.\chi$ embeds an arbitrary proposition χ into this mechanism. Together with T3, which supplies necessarily existing God-like witnesses, and suitable frame conditions, the standard Scott setting yields

$$\chi \rightarrow \Box \chi.$$

The detailed classical dependency analysis is maintained in `docs/MODAL_COLLAPSE_SPINE.md`. This local structure is consonant with the recent machine-checked diagnosis that positivity rigidity, rather than ultrafilter structure alone, is a principal driver of collapse [2].

6.1 Correct bilateral collapse targets

The informative signed persistence principles are

$$\text{MC}^+ : \quad +\chi \Rightarrow +\Box \chi$$

and

$$\text{MC}^- : \quad -\chi \Rightarrow +\Box \neg \chi.$$

The earlier candidate $-\chi \Rightarrow -\Box \chi$ is discarded because it follows automatically on reflexive frames: the current world itself witnesses negative satisfaction of $\Box \chi$.

Theorem 6.1 (Collapse-channel equivalence). *Under the fixed FDE negation, with both schemata ranging over all formulas,*

$$\text{MC}^+ \iff \text{MC}^-.$$

Proof. Assume MC^+ and $\neg\chi$. By FDE negation, $+\neg\chi$. Instantiating MC^+ with $\neg\chi$ gives $+\Box\neg\chi$, hence MC^- .

Conversely, assume MC^- and $+\chi$. Then $\neg\neg\chi$. Instantiating MC^- with $\neg\chi$ yields $+\Box\neg\neg\chi$, which is equivalent to $+\Box\chi$ because FDE negation is involutive. $\square$

Thus the original conjecture that the informative positive and negative collapse channels might separate is falsified for the present semantics. The decomposition must be sought upstream rather than forced at the collapse schema itself.

6.2 Rigidity separates before A1 couples it

Write

$$R^+ : \quad +P(\varphi) \Rightarrow +\Box P(\varphi)$$

and

$$R_{\text{nec}}^- : \quad -P(\varphi) \Rightarrow +\Box\neg P(\varphi).$$

Without A1 these principles are independent. On the two-world S5 frame with universal accessibility, assign a property ψ the positivity values

$$v_{w_0}(P(\psi)) = \mathbf{N}, \quad v_{w_1}(P(\psi)) = \mathbf{F}.$$

If all other relevant positivity statements receive $\mathbf{N}$, then R^+ holds vacuously while R_{nec}^- fails at w_1 . Replacing $\mathbf{F}$ by $\mathbf{T}$ yields the converse separation: R_{nec}^- holds vacuously while R^+ fails.

Strong A1 destroys this independence.

Proposition 6.2 (Rigidity coupling under strong A1). *Under $A1_L + A1_R$,*

$$R^+ \iff R_{\text{nec}}^-.$$

Proof. The direction $R^+ \Rightarrow R_{\text{nec}}^-$ was established in the positivity analysis. For the converse, suppose $+P(\varphi)$. By $A1_R$, instantiated at $\neg\varphi$, one obtains $-P(\neg\varphi)$. Applying R_{nec}^- to $\neg\varphi$ yields $+\Box\neg P(\neg\varphi)$. At each accessible world, $A1_L$ converts $-P(\neg\varphi)$ back into $+P(\varphi)$. Hence $+\Box P(\varphi)$. $\square$

The genuine bilateral separation therefore occurs at the level of positivity rigidity only in the absence of the full A1 coupling.

6.3 Minimal recovery of the local reflection bridge

The signed reflection target is

$$REF^+ : \quad +G(x) \wedge +Z(x) \Rightarrow +P(Z).$$

Strong $A1$ plus the minimal support interface

$$D1^+ : \quad +G(x) \wedge +P(\varphi) \Rightarrow +\varphi(x)$$

does not suffice: the glut and gap assignments analyzed in Section 5 block the classical reductio in different ways.

Introduce two explicit metalanguage regularity conditions for the relevant witness and property:

$$CONS_G : \quad +G(x) \wedge +Z(x) \Rightarrow \text{not } -Z(x),$$

and

$$COMP_P : \quad +P(Z) \text{ or } -P(Z).$$

Proposition 6.3 (Local reflection recovery). *At the abstract Gate-4 interface,*

$$A1_L + D1^+ + CONS_G + COMP_P \models REF^+.$$

Proof. Assume $+G(x)$ and $+Z(x)$. By $COMP_P$, either $+P(Z)$ or $-P(Z)$. In the latter case, $A1_L$ yields $+P(\neg Z)$, and $D1^+$ then gives $+\neg Z(x)$, equivalently $-Z(x)$. This contradicts $CONS_G$. Hence the negative-positivity alternative is excluded and $+P(Z)$ remains. $\square$

The package is locally minimal relative to this interface vocabulary. Removing $A1_L$, $D1^+$, $CONS_G$, or $COMP_P$ respectively admits a counterassignment while the other three remain satisfiable. The glut counterassignment witnesses the need for $CONS_G$; the gap counterassignment witnesses the need for $COMP_P$.

6.4 Conditional collapse kernel

Two higher-order interfaces are kept explicit because the final quantifier and Godlikeness semantics have not yet been frozen:

- GW : every world has some positively supported God-like witness;
- $CONST$: every modal formula χ is representable by the constant intensional property $Z_\chi := \lambda x.\chi$, preserving both signed satisfaction conditions.

Theorem 6.4 (Interface-level collapse package). *Assume globally*

$$A1_L + D1^+ + CONS_G + COMP_P + R^+ + GW + CONST.$$

Then MC^+ , and therefore also MC^- .

Proof. Suppose $+\chi$ at w . By *GW*, choose x with $+G(x)$ at w . By *CONST*, $+Z_\chi(x)$. The local reflection proposition yields $+P(Z_\chi)$. By R^+ , $+\Box P(Z_\chi)$.

Let v be accessible from w . Then $+P(Z_\chi)$ at v . By *GW*, choose y with $+G(y)$ at v . By $D1^+$, $+Z_\chi(y)$, and by *CONST*, $+\chi$ at v . Since v was arbitrary, $+\Box\chi$ at w . $\square$

This is deliberately an interface-level theorem rather than a theorem of the complete four-valued Gödel–Scott theory. Gate 5 must determine whether the final treatment of A2, A3, D1, and higher-order quantification justifies *GW* and *CONST*, and whether the regularity assumptions can be weakened or derived rather than postulated.

Gate-4 conclusion. The strongest current structural claim is therefore not that modal collapse itself splits. Under symmetric FDE negation it does not. The non-classical decomposition occurs upstream: positivity rigidity separates before strong A1 recouples it, while the classical reflection bridge additionally relies on consistency and completeness conditions that gluts and gaps defeat independently.

7 Reconstructing the Gödel–Scott Chain

Gates 5 and 6 reconstruct the higher-order existence branch through A2, A3, Godlikeness, essence, necessary existence, A5, and a positive analogue of Scott’s T3. The reconstruction keeps the object-language four-valued conditional unfixed; implication-sensitive clauses are represented semantically.

7.1 Actualist quantification and necessary positive entailment

I retain Scott’s actualist quantification over individuals using a classical metalanguage existence relation $E(w, x)$. Positive and negative existential satisfaction are

$$w \models^+ \exists^E x \varphi(x) \iff \exists x (E(w, x) \wedge w \models^+ \varphi(x)),$$

$$w \models^- \exists^E x \varphi(x) \iff \forall x (E(w, x) \Rightarrow w \models^- \varphi(x)),$$

with the dual universal clauses.

Necessary positive entailment is the semantic relation

$$\text{NEnt}_E^+(\varphi, \psi, w),$$

which holds iff at every world accessible from w , every actually existing positive instance of φ is a positive instance of ψ . The control lifting of A2 is

$$A2^+ : \quad +P(\varphi) \wedge \text{NEnt}_E^+(\varphi, \psi) \Rightarrow +P(\psi).$$

For Gate 6 I also use negative evidence against necessary entailment:

$$\text{NEnt}_E^-(\varphi, \psi, w)$$

iff some accessible actually existing object positively exemplifies φ and negatively exemplifies ψ . On the classical fragment these two relations reproduce truth and falsity of the necessary-entailment condition.

7.2 Truth-only positivity and possible exemplification

Define

$$\text{Pos}_T(\varphi) \iff +P(\varphi) \text{ and not } -P(\varphi),$$

where “not” is metalanguage denial.

Theorem 7.1 (Truth-only positive properties are possibly exemplified). *Assume globally $A1_R$ and $A2^+$. Then*

$$\text{Pos}_T(\varphi) \Rightarrow +\Diamond\exists^E x \varphi(x).$$

The proof is the Gate-5 reductio: failure of positive possible exemplification makes $\text{NEnt}_E^+(\varphi, \neg\varphi)$ vacuously true; $A2$ then gives $+P(\neg\varphi)$, and $A1_R$ yields $-P(\varphi)$, contradicting truth-only positivity. A one-world model with an empty actual domain and gluttony positivity shows that the stronger antecedent $+P(\varphi)$ is insufficient.

Scott $A3$ is therefore split into

$$A3^+ : +P(G), \quad A3_T : \text{Pos}_T(G).$$

The control branch uses $A3_T$, obtaining

$$\boxed{+\Diamond\exists^E x G(x)}.$$

7.3 Support-based Godlikeness

The control definition $G\text{-sup-v0.1}$ is

$$\begin{aligned} w \models^+ G(x) &\iff \forall\varphi (w \models^+ P(\varphi) \Rightarrow w \models^+ \varphi(x)), \\ w \models^- G(x) &\iff \exists\varphi (w \models^+ P(\varphi) \wedge w \models^- \varphi(x)). \end{aligned}$$

This recovers Scott $D1$ on the classical fragment, discharges the former $D1^+$ interface, and permits all four information states for Godlikeness. Full higher-order lambda abstraction supplies the constant-property interface $CONST$.

7.4 Bilateral essence

The Gate-6 control definition *Ess*-sup-v0.1 is

$$w \models^+ (\varphi \text{Ess } x) \iff w \models^+ \varphi(x) \wedge \forall \psi (w \models^+ \psi(x) \Rightarrow \text{NEnt}_E^+(\varphi, \psi, w)),$$

while

$$w \models^- (\varphi \text{Ess } x) \iff w \models^- \varphi(x) \vee \exists \psi (w \models^+ \psi(x) \wedge \text{NEnt}_E^-(\varphi, \psi, w)).$$

On the classical fragment these clauses recover Scott D2.

The positive T2 target is

$$T2^+ : \quad +G(x) \Rightarrow +G \text{Ess } x.$$

T2 is not automatic. Two explicit two-world S5 models refute $T2^+$ while satisfying the Gate-5 control stack together with strong A1 and R^+ . In the glut model, a property Z is both positively and negatively exemplified by one positive God-like witness but lacks positive support at another accessible God-like witness. In the gap model, $P(Z) = P(\neg Z) = \mathbf{N}$, while Z is positively exemplified locally but not necessarily shared. In both cases $A2^+$, $A3_T$, G -sup-v0.1, strong A1, and positive rigidity hold, yet $\text{NEnt}_E^+(G, Z)$ fails. Hence positive Godlikeness does not imply positive Godlikeness-as-essence merely by choosing a natural bilateral D2.

7.5 A sufficient recovery route for T2

For properties actually positively exemplified by positive God-like witnesses, define

$$\text{COMP}_P^G : \quad +G(x) \wedge +\psi(x) \Rightarrow (+P(\psi) \text{ or } -P(\psi)),$$

and

$$\text{CONS}_G^G : \quad +G(x) \wedge +\psi(x) \Rightarrow \text{not } -\psi(x).$$

Let REG_G be their conjunction.

Theorem 7.2 (Conditional recovery of positive T2). *Under $A1_L$, R^+ , G -sup-v0.1, and REG_G ,*

$$T2^+$$

holds.

Proof. Assume $+G(x)$ and let ψ be positively exemplified by x . Relevant positivity completeness gives $+P(\psi)$ or $-P(\psi)$. The latter alternative would, by $A1_L$, make $\neg\psi$ positively positive; Godlikeness would then yield $-\psi(x)$, contradicting relevant exemplification consistency. Hence $+P(\psi)$. Positive

rigidity makes this support necessary, and every accessible positive God-like witness consequently positively exemplifies ψ . Thus $\text{NEnt}_E^+(G, \psi)$, as required by positive essence. $\square$

The glut and gap countermodels isolate the two regularity failures for this recovery route. I do not claim that REG_G is globally necessary for every model satisfying $T2^+$.

7.6 Bilateral necessary existence and A5

Define $NE\text{-sup-v0.1}$ by

$$\begin{aligned} w \models^+ NE(x) &\iff \forall \varphi (w \models^+ (\varphi \text{ Ess } x) \Rightarrow w \models^+ \Box \exists^E y \varphi(y)), \\ w \models^- NE(x) &\iff \exists \varphi (w \models^+ (\varphi \text{ Ess } x) \wedge w \models^- \Box \exists^E y \varphi(y)). \end{aligned}$$

These clauses recover Scott D3 classically.

For A5 distinguish

$$A5^+ : \quad +P(NE), \quad A5_T : \quad \text{Pos}_T(NE).$$

Unlike A3, the positive T3 branch needs only $A5^+$. Once $+G(x)$ is available, $G\text{-sup-v0.1}$ turns $+P(NE)$ directly into $+NE(x)$. Truth-only positivity of NE is unnecessary for that positive inference.

7.7 Positive necessary God-like existence

Theorem 7.3 (Positive T3 with symmetry). *Assume positive possible God-like existence, $T2^+$, $A5^+$, $G\text{-sup-v0.1}$, $NE\text{-sup-v0.1}$, and symmetry of the accessibility relation. Then*

$$T3^+ : \quad +\Box \exists^E x G(x).$$

No reflexivity or transitivity premise is required by this T3 route.

Proof. Fix a world w and an arbitrary successor z of w . Positive possible God-like existence supplies a world v accessible from w and an actual positive God-like witness x at v . By A5 and Godlikeness, x positively exemplifies NE ; by T2, G is a positive essence of x . Hence $NE\text{-sup-v0.1}$ supplies an actual positive G -instance at every successor of v .

By symmetry, wRv gives vRw , so one such actual positive G -instance exists already at w . The G -realization turns it into a positive God-like witness at w . Applying A5, T2, and $NE\text{-sup-v0.1}$ once more at w supplies an actual positive G -instance at the arbitrary successor z . Thus every successor of w has an actual positive God-like witness, establishing $T3^+$. $\square$

A finite two-world countermodel shows that reflexivity plus transitivity do not replace symmetry for this route. The frame $w_0Rw_0, w_0Rw_1, w_1Rw_1$ satisfies the encoded Scott premises of the reduced theorem other than symmetry, including $T2^+$, $A5^+$, NE -realization, G -realization, and possible Godlikeness, while $T3^+$ fails at w_0 .

On reflexive frames, $T3^+$ additionally supplies an actually existing positive God-like witness at every world, so the former interface GW is discharged. Reflexivity is therefore needed for this particular $T3^+ \Rightarrow GW$ step, not for the symmetry-only T3 theorem itself.

Combining the sufficient T2 route with Gate 5 gives the conditional positive Scott-control chain

$$A1_{\leftrightarrow} + A2^+ + A3_T + R^+ + REG_G + A5^+ + \text{Sym}(R) \Longrightarrow T3^+,$$

under the fixed higher-order semantics.

7.8 Essence-compressed modal collapse

Once $T2^+$ and $T3^+$ are available, the earlier reflection/rigidity proof can be compressed.

Theorem 7.4 (Essence-compressed collapse). *Under $T2^+$, $T3^+$, $G\text{-sup-v0.1}$, constant-property embedding, and reflexivity of the accessibility relation,*

$$\text{MC}^+$$

holds, and therefore MC^- holds by the Gate-4 collapse-equivalence theorem.

Proof. Assume $+\chi$ at w . Reflexivity together with T3 supplies an actual positive God-like witness x at w , and T2 makes G a positive essence of x . For the constant property $Z_\chi = \lambda y. \chi, +Z_\chi(x)$. Positive essence gives $\text{NEnt}_E^+(G, Z_\chi, w)$. T3 supplies a positive God-like witness at every accessible world; $G\text{-sup-v0.1}$ identifies each such witness with a positive G -instance. Necessary entailment therefore makes each such witness positively exemplify Z_χ , hence each accessible world positively supports χ . Therefore $+\Box\chi$. $\square$

This result refines the Gate-4 dependency map. The earlier REF^+ , $CONS_G$, $COMP_P$, and R^+ assumptions are ingredients of one route for deriving T2 from lower-level principles; they are not separate premises of the essence-compressed collapse theorem after T2 and T3 have already been established.

7.9 Robustness and the new bottleneck

A5 illustrates genuine inconsistency tolerance. If $P(NE) = \mathbf{B}$, strong A1 also permits $P(\neg NE) = \mathbf{B}$. A positive God-like witness may therefore satisfy

both $+NE(x)$ and $-NE(x)$, while the positive component remains sufficient for the T3 derivation.

Likewise, global positivity completeness and global consistency of arbitrary God-like exemplification are not semantic prerequisites for every model satisfying the positive T2/T3 chain. They are route-specific regularity assumptions used to recover T2 from the primitive lower-level axioms.

The decisive open question after Gate 6 is consequently not whether necessary existence can supply witnesses. It can, once T2 is available. The main remaining semantic-minimality problem is:

What are the weakest principled four-valued conditions under which $T2^+$ follows?

The full Gate-5 and Gate-6 specifications are maintained in `docs/GODLIKENESS_AND_POSSIBILITY`. and `docs/ESSENCE_AND_NECESSARY_EXISTENCE.md`.

8 Mechanization and Countermodels

STATUS: ESTABLISHED AT THE SEMANTIC LEVEL.

Gate 7 adds two complementary verification layers. I use a finite-model oracle for concrete countermodels and bounded exhaustive searches, and Lean for general semantic theorems that must not depend on a finite bound. Gate 8 extends the same discipline to comparative variants, modal-frame minimization, and the Fitting intension/extension reconstruction. This separation prevents the absence of a small countermodel from being mistaken for a proof and, conversely, allows an apparently successful proof interface to be tested for semantic vacuity.

8.1 Finite-model regression layer

The executable checker implements the four information values, involutive FDE negation, finite accessibility relations, actualist existence, four-valued property extensions and positivity, signed necessary entailment, support-based Godlikeness, positive essence, and the Gate-6 regularity conditions.

The regression suite machine-checks both two-world T2 countermodels from Gate 6. The glut model satisfies the encoded Gate-5 control stack and $COMP_P^G$ but violates $CONS_G^G$, while the gap model satisfies the same control stack and $CONS_G^G$ but violates $COMP_P^G$. Both refute $T2^+$.

The checker also validates the Gate-5 obstruction to unrestricted positive T1. A one-world reflexive model with no actual entities and glutty positivity for a property and its complement satisfies strong A1 and $A2^+$, while positive support for the property does not yield positive possible actual exemplification. This is the executable counterpart of the distinction between mere positive support and truth-only positivity.

A first bounded search retains 204 models satisfying the original small control antecedent and verifies $T2^+$ in every retained model. A broader exhaustive two-world, one-entity G, Z family then varies positivity of Z and its complement independently while fixing FDE complement exemplification and full bilateral G -support semantics. In this broader family, 873 models satisfy

$$A1_L, \quad R^+, \quad COMP_P^G, \quad CONS_G^G,$$

and every one of them satisfies $T2^+$. Dropping any one of these four assumptions while retaining the other three produces a $T2^+$ countermodel within the same bounded family. I therefore treat this as bounded individual indispensability, not global model-theoretic minimality.

The same oracle now exports selected witnesses as directed Kripke graphs. Figure 1 shows the first countermodel isolated after dropping only $A1_L$: R^+ , $COMP_P^G$, and $CONS_G^G$ remain true while $T2^+$ fails. The two HTML-style node tables display the complete local FDE matrix. Each row records its value in $\{\mathbf{T}, \mathbf{F}, \mathbf{B}, \mathbf{N}\}$ together with the underlying positive and negative support bits. The four arrows are read directly from the model's complete two-world accessibility relation.

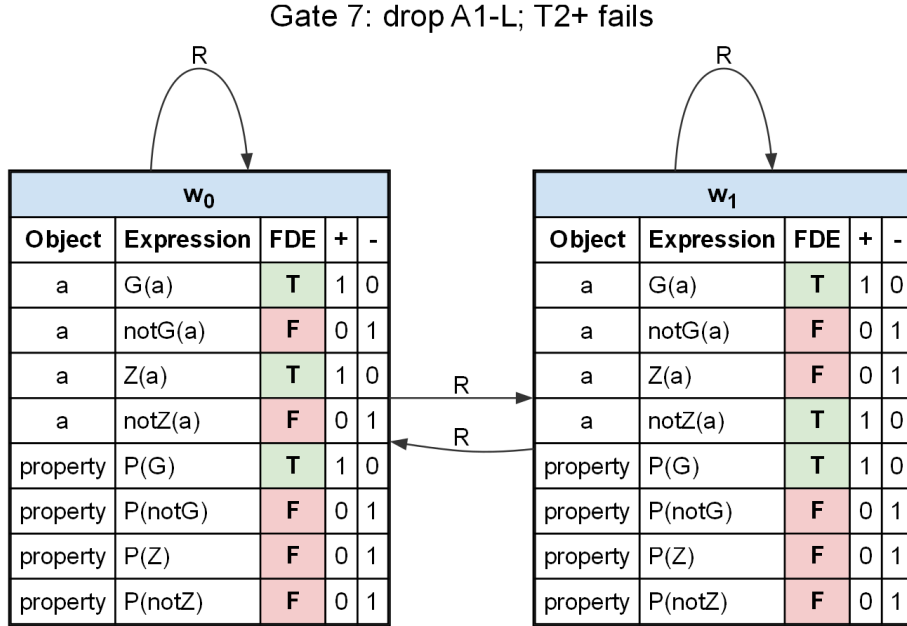


Figure 1: The first Gate-7 $T2^+$ countermodel found when only $A1_L$ is dropped. The graph is generated by `formal/finite/visualize_gate7_models.py` from the same matrix family used in the 873-model exhaustive regression.

The checker also verifies the schema-level equivalence

$$MC^+ \leftrightarrow MC^-$$

on a negation-closed two-world test family. The equivalence concerns schemata closed under substitution by negation, not a pointwise identity for one isolated formula valuation.

Gate 8 adds comparative witnesses. The support/exact comparison confirms that both Gate-6 T2 countermodels are support-Godlike but fail project-internal exact positive Godlikeness, while a separate exact-Godlike fixture still permits genuine **B** gluts. A complete-S5 bilateral Anderson fixture has positive necessary Godlike existence while a contingent formula application refutes positive modal collapse. Separate Scott and Anderson S4-style two-world models are reflexive and transitive but not symmetric and refute the corresponding reduced T3 conclusions.

The Fitting branch adds two further finite witnesses. First, a three-world two-entity model uses a selected negation-closed extensional property domain and satisfies the admissible Fitting recovery stack. At the root world the current frozen G -extension is both possibly and necessarily actually exemplified de re, and de-dicto Godlikeness is possible, but de-dicto necessary Godlikeness fails because the extension of G changes at another accessible world. The model therefore separates the two readings when the explicit extension-stability condition introduced below is absent.

Second, a complete two-world S5 model satisfies the encoded admissible Fitting stack, extension stability, and positive necessary actual Godlikeness. Its selected extension domain nevertheless contains genuine inconsistent information: one admissible extension takes value **B** on an entity, together with its FDE negation. A separate contingent intension Q satisfies

$$+Q(a)@w_0$$

but not

$$+\Box Q(a)@w_0.$$

Thus the present admissible bilateral Fitting candidate has an explicit necessary-God / no-positive-collapse model. This is a finite countermodel to collapse for the candidate, not an unbounded theorem about every Fitting semantics.

8.2 Lean layer

The Lean package is pinned to Lean 4.30.0. It formalizes the four-valued carrier, FDE negation, conjunction and disjunction, bilateral relational modality, bilateral actualist quantification, signed necessary entailment, support-based Godlikeness, bilateral essence, and necessary-existence interfaces.

Lean proves the collapse-channel theorem generally at schema level and the Gate-5 truth-only possibility theorem

$$A1_R + A2^+ \implies T1_T.$$

The principal Gate-6 Scott recovery theorem is machine-checked without a finite bound:

$$NegExemplification + G\text{-sup} + A1_L + R^+ + REG_G \Longrightarrow T2^+.$$

Gate 8 sharpens the later modal dependency. Once Scott $T2^+$ is available, Lean proves

$$\text{Sym}(R) + PossibleGod + T2^+ + A5^+ + NE\text{-sup} + G\text{-sup} \Longrightarrow T3^+.$$

Reflexivity and transitivity are absent. Reflexivity enters only in the separate implication $T3^+ \Rightarrow GW$ and in the current essence-compressed collapse theorem

$$\text{Refl}(R) + T2^+ + T3^+ + G\text{-sup} + CONST \Longrightarrow MC^+.$$

For project-internal exact positive Godlikeness, Lean proves

$$G\text{-exact}^+ \Rightarrow G\text{-sup}^+$$

and

$$G\text{-exact}^+ + R^+ \Longrightarrow T2_{\text{exact}}^+.$$

For Anderson, the literature-grounded positive interfaces and project-specific bilateral negative evidence are mechanized separately. Under classical coherence the negative necessary-exemplification, Godlikeness, essence, necessary-actual-exemplification, and necessary-existence clauses recover the complements of their positive counterparts. The positive Anderson necessary-existence branch reduces to symmetry alone:

$$\begin{aligned} &\text{Sym}(R) + PossibleG_A + G_A\text{-realization} + P^+(G_A) + A2^+ \\ &\quad + NE_A\text{-realization} + P^+(NE_A) \Longrightarrow T3_A^+. \end{aligned}$$

8.3 Fitting: from a type distinction to a domain obstruction

The Fitting comparison begins by distinguishing bilateral extensions from intensions:

$$Extension(Entity) \quad \text{and} \quad Intension(World, Entity) = World \rightarrow Extension(Entity).$$

The positivity predicate is typed over extensions, and *extensionAt* makes the passage from an intension to its extension at a world explicit. Signed necessary entailment, essence, and necessary existence are then reconstructed over rigid extensions, reflecting the extensional architecture of the mechanized Fitting variant [3].

A naive first lift quantified Fitting Godlikeness and the relevant consistency/completeness package over every inhabitant of the bilateral extension

type. Lean exposes a problem with that choice. The unrestricted type contains the extension B satisfying

$$+B(x) \quad \text{and} \quad -B(x)$$

for every individual. Consequently Lean proves

$$CONS_{G,F}^{\text{all}} \implies \forall w \forall x \neg G_F^+(x, w),$$

and therefore also

$$REG_{G,F}^{\text{all}} \implies \forall w \forall x \neg G_F^+(x, w).$$

The earlier unrestricted implication from this regularity package to extensional essence remains a valid implication, but its intended Godlike antecedent is excluded by the package itself. I therefore do not count that unrestricted route as the substantive Fitting recovery result. The failed modeling choice is retained in the formal development because the obstruction is itself informative.

8.4 Admissible Fitting extensions

The corrected candidate uses a selected domain of admissible extensions, written $Adm(X)$, with FDE-negation closure

$$Adm(X) \implies Adm(\neg X).$$

This is a property-domain restriction rather than a global consistency postulate. In particular, admissible extensions may remain gluttony, as the finite S5 witness above demonstrates.

Positive Fitting Godlikeness becomes

$$G_F^+(x, w) \quad \text{iff} \quad \forall X (Adm(X) \wedge +P(X)@w \Rightarrow +X(x)).$$

The relevant Fitting regularity package is likewise restricted to admissible extensions. Lean then proves the non-vacuous extensional essence theorem

$$\begin{aligned} & G\text{-admissible} + G_F\text{-realization} + A1_{L,F}^{\text{adm}} + REG_{G,F}^{\text{adm}} \\ & \implies (G_F^+(x, w) \Rightarrow Ess_F^+(ext_w G, x, w)). \end{aligned}$$

Unlike the current Scott-support T2 recovery theorem, no R^+ premise occurs. Once local positivity of an admissible rigid extension has been recovered, its membership relation does not itself vary across accessible worlds.

The bilateral negative evidence is also audited. Under classical coherence on the selected extension domain, Lean proves the recovery schemata

$$\begin{aligned} Ent_F^- & \leftrightarrow \neg Ent_F^+, \\ G_F^- & \leftrightarrow \neg G_F^+, \\ Ess_F^- & \leftrightarrow \neg Ess_F^+, \\ BoxExists_F^- & \leftrightarrow \neg BoxExists_F^+, \\ NE_F^- & \leftrightarrow \neg NE_F^+. \end{aligned}$$

These results validate the project-specific negative clauses at the classical boundary without imposing classicality on the four-valued candidate itself.

8.5 Fitting de re and de dicto

Let $ext_w G$ denote the extension of the Godlikeness intension frozen at the evaluation world. With a distinguished NE intension realizing admissible-domain necessary existence and positive A5 on its current extension, Lean proves

$$\begin{aligned} & \text{admissible Fitting stack} + \Diamond_{dere}^+ \exists^E x ext_w G(x) \\ & \implies \Box_{dere}^+ \exists^E x ext_w G(x). \end{aligned}$$

No reflexivity, symmetry, transitivity, seriality, or positivity-rigidity premise occurs. This is the project's four-valued counterpart of the K-level de-re behavior emphasized in the mechanized Fitting analysis [3].

The de-dicto claim is intentionally not identified with this theorem. It re-evaluates G at each accessible world. I isolate the required intension/extension bridge as

$$STAB_G : \quad wRz \Rightarrow ext_w G \equiv ext_z G.$$

Lean proves that $STAB_G$ converts de-dicto possibility to the frozen de-re premise and converts the de-re necessary conclusion back to de-dicto necessary Godlikeness. Hence

$$\begin{aligned} & \text{admissible Fitting stack} + STAB_G + \Diamond^+ \exists^E x G_F(x) \\ & \implies \Box^+ \exists^E x G_F(x), \end{aligned}$$

again without an S4 or S5 frame premise. The three-world finite witness shows that the stability condition is not merely decorative for this proof interface.

The minimized positive bridge admits a structural formulation. Lean defines $R_{F,adm}^+$ as forward positivity rigidity restricted to admissible rigid extensions and $R_{F,adm}^{+\leftarrow}$ as converse positivity transport along existing accessibility edges. It proves

$$\begin{aligned} R_{F,adm}^+ + G_F\text{-realization} & \implies Refl_G^+, \\ R_{F,adm}^{+\leftarrow} + G_F\text{-realization} & \implies Pers_G^+, \\ R_{F,adm}^+ + Sym(R) & \implies R_{F,adm}^{+\leftarrow}. \end{aligned}$$

Consequently $R_{F,adm}^+ + R_{F,adm}^{+\leftarrow}$ can replace primitive $STAB_G^+$ in the positive de-dicto theorem without any frame condition. The finite oracle checks this bridge exhaustively over 256 two-world, one-entity configurations in the classical negation-closed admissible fragment. Among 112 candidates satisfying both transports, 32 are non-symmetric; converse positivity transport is therefore strictly weaker than symmetry in the tested family.

8.6 Classical recovery and verification discipline

The Scott, Anderson, and admissible Fitting branches each have explicit recovery interfaces rather than relying on an informal assertion that the bilateral semantics “becomes classical.” Exact higher-order correspondence with the complete AFP/HOL developments is nevertheless a separate publication audit item.

The repository treats a countermodel or an obstruction theorem as a result rather than as a failed proof attempt. The unrestricted Fitting-comprehension obstruction is a particularly useful example: a theorem initially looked stronger because it lacked R^+ , but mechanization plus domain analysis revealed that the unrestricted regularity package excluded its intended witnesses. The admissible-domain reconstruction makes the subsequent theorem non-vacuous and is independently checked by finite models.

All executable and Lean sources are maintained under `formal/`, with CI configured to run both tracks. General claims are promoted to machine-checked status only after an unbounded Lean theorem is present, while finite witnesses are reported as finite witnesses with their model bounds explicit.

8.7 Fitting assumption minimization

The admissible-domain reconstruction permits a second round of dependency analysis. The original recovery package

$$REG_{G,F}^{adm} = COMP_{P,F}^{G,adm} + CONS_{G,F}^{G,adm}$$

requires every admissible extension positively possessed by a positive Fitting-Godlike witness to be locally non-glutty. The proof of the extensional essence theorem, however, uses this consistency information only when positivity completeness classifies the possessed extension negatively. I therefore isolate the weaker condition

$$\begin{aligned} NegCons_{G,F}^{adm} : \quad & Adm(Y) \wedge G_F^+(x, w) \wedge +Y(x) \wedge -P(Y)@w \\ & \implies \neg -Y(x). \end{aligned}$$

Writing

$$REG_{G,F}^{neg} := COMP_{P,F}^{G,adm} + NegCons_{G,F}^{adm},$$

Lean proves

$$CONS_{G,F}^{G,adm} \implies NegCons_{G,F}^{adm}$$

and the strengthened recovery theorem

$$\begin{aligned} & G\text{-admissible} + G_F\text{-realization} + A1_{L,F}^{adm} + REG_{G,F}^{neg} \\ & \implies G_F^+(x, w) \Rightarrow Ess_F^+(ext_w G, x, w). \end{aligned}$$

The corresponding de-re necessary-existence chain is machine-checked with the same targeted package. A one-world one-entity finite model separates the two consistency notions on the A1-L-only route: the sole admissible extension is genuinely glutty, the targeted negative-classification condition holds, full relevant exemplification consistency fails, and the realized positive Godlike extension is nevertheless an essence.

This weakening is itself directional. If the right A1 channel is restored, Lean proves

$$A1_{R,F}^{adm} + COMP_{P,F}^{G,adm} + NegCons_{G,F}^{adm} \implies CONS_{G,F}^{G,adm}.$$

Thus the reduction is not an unconditional weakening of a theory retaining both A1 directions. It instead exposes another division of labor between the two signed A1 flows: the A1-L recovery argument needs consistency only on the negative-classification branch, while A1-R supplies enough additional complement information to reconstruct the stronger consistency condition.

Why completeness cannot simply be renamed away. Inside this classification-based proof, $COMP_{P,F}^{G,adm}$ supplies exactly the gap branch: a relevant extension either already has positive positivity support or is negatively classified so that A1-L can activate the complement route. Replacing the disjunction by a condition of the form

$$\neg +P(Y)@w \implies -P(Y)@w$$

would therefore not constitute a genuine weakening at the classical meta-level; on the same relevant domain it is equivalent to the branch coverage already expressed by completeness. A real reduction of $COMP$ must consequently change the recovery mechanism itself.

An alternative recovery route. I introduce a second, deliberately non-classificatory interface. Let admissible-property indiscernibility among positive Fitting-Godlike individuals be

$$\begin{aligned} Ind_G^{adm} : & \quad Adm(Y) \wedge G_F^+(x, w) \wedge G_F^+(y, w) \\ & \implies (+Y(x) \leftrightarrow +Y(y)). \end{aligned}$$

This condition does not assign positive or negative positivity status to Y . Instead, it constrains the selected extensional property domain: admissible rigid extensions cannot positively distinguish two positive Godlike individuals at the same source world.

Lean proves

$$G\text{-admissible} + G_F\text{-realization} + Ind_G^{adm} \implies G_F^+(x, w) \Rightarrow Ess_F^+(ext_w G, x, w).$$

No A1 direction, no positivity completeness, and no exemplification-consistency premise occurs. The proof is extensional. If Y contains the selected Godlike individual x , any entity y in the frozen current G -extension is also positive Godlike at the source world by G -realization. Indiscernibility then transfers positive Y -membership from x to y , yielding exactly the rigid necessary entailment required by Fitting essence.

The same route continues through necessary existence. Lean machine-checks a frame-free de-re possibility-to-necessity theorem from G -admissibility, G -realization, Ind_G^{adm} , NE-admissibility, NE-realization, and positive A5, with no $COMP$, A1, or consistency premise. Positive G -extension persistence and reflection then provide the same de-dicto bridge as before.

Positive-profile quotient structure. The indiscernibility premise can itself be derived from a more structural condition on the selected property domain. Define positive-profile equivalence at a world by

$$x \approx_w^+ y$$

iff every positively supported admissible extension contains x and y alike. Lean proves that $\approx_w^+$ is reflexive, symmetric, and transitive. Positive Fitting-Godlike individuals automatically lie in the same $\approx_w^+$ -class, because each of them positively exemplifies every positively supported admissible extension.

I then impose the quotient-style saturation condition

$$\begin{aligned} Sat_P^{adm} : \quad & Adm(Y) \wedge x \approx_w^+ y \\ & \implies (+Y(x) \leftrightarrow +Y(y)). \end{aligned}$$

Lean proves

$$Sat_P^{adm} \implies Ind_G^{adm},$$

and therefore the entire alternative essence/NE route follows from this property-domain condition. This gives the previously primitive indiscernibility interface a quotient-semantical interpretation rather than treating it as an isolated recovery axiom.

The quotient is in fact bilateral. Since the substantive admissible semantics is already closed under FDE negation, applying Sat_P^{adm} to $\neg Y$ transfers constancy of positive membership into constancy of negative membership of Y . Lean therefore proves

$$\boxed{Sat_P^{adm} \iff Sat_{\pm}^{adm}},$$

where $Sat_{\pm}^{adm}$ requires both support channels of every admissible extension to be constant on the same positive-profile classes. Thus each admissible bilateral extension genuinely factors through one entity quotient rather than merely sharing a positive projection.

A one-world, three-entity finite fixture makes the result nontrivial. Two distinct entities a and b are both positive Godlike and form one non-singleton positive-profile class, while c is not Godlike. The negation-closed admissible domain consists of the rigid extensions

$$\emptyset, \quad \{a, b\}, \quad \{c\}, \quad \{a, b, c\}.$$

Every admissible extension respects the $\{a, b\}$ profile class, so Sat_P^{adm} , its bilateral counterpart, and Ind_G^{adm} hold. Nevertheless the universal extension is a genuine positivity gap:

$$\neg +P(\{a, b, c\}) \quad \text{and} \quad \neg -P(\{a, b, c\}),$$

so $COMP_{P,F}^{G,adm}$ fails at an extension containing both Godlike individuals. The current G -extension $\{a, b\}$ is still an essence of each, and the finite NE/A5 interface remains satisfied. Hence the profile-saturation route bypasses positivity completeness non-vacuously even when Godlike indiscernibility has real two-individual content.

Algebraic stability of the quotient. The quotient interpretation is compatible with the basic FDE property operations. At the extension level define conjunction and disjunction pointwise by

$$\begin{aligned} +(X \wedge_F Y)(x) &\Leftrightarrow +X(x) \wedge +Y(x), & -(X \wedge_F Y)(x) &\Leftrightarrow -X(x) \vee -Y(x), \\ +(X \vee_F Y)(x) &\Leftrightarrow +X(x) \vee +Y(x), & -(X \vee_F Y)(x) &\Leftrightarrow -X(x) \wedge -Y(x). \end{aligned}$$

Lean proves that, for each fixed positive-profile quotient, the bilateral extensions that factor through it contain the FDE top and bottom extensions and are closed under FDE negation, conjunction, and disjunction. In this precise preservation sense the quotient-respecting extensions form a De-Morgan-style subalgebra of the full bilateral extension algebra.

This theorem does not itself impose algebraic closure on Adm . It establishes the weaker and methodologically prior fact that adding FDE compounds of quotient-respecting extensions cannot break the quotient factorization. Whether the intended Fitting property domain should actually be closed under all of these operations remains a separate semantic choice.

Relation to Fitting's ultrafilter structure. The published mechanized analysis proves that Fitting's extensional positivity predicate constitutes a δ -ultrafilter [3]. Its classical maximality component chooses an extension or its complement as positive. To compare that architecture with the bilateral reconstruction, I isolate a four-valued complement-decision interface:

$$Adm(Y) \implies (+P(Y) \vee +P(\neg Y)).$$

This condition alone is not yet a four-valued ultrafilter definition. More importantly, the bilateral setting changes what the complement branch

supplies: positive support for $\neg Y$ yields negative support information about membership in Y , but it does not by itself erase positive support for Y .

Lean shows that complement decision combined with full relevant exemplification consistency already recovers local positive reflection at Godlike witnesses, and hence $COMP_{P,F}^{G,adm}$. Independently, complement decision together with $A1_R$ also yields $COMP_{P,F}^{G,adm}$. Thus the ultrafilter-style route explains why completeness is natural at the classical Fitting boundary, but it does not account for the genuinely classification-free route above. The quotient-saturation condition and ultrafilter maximality therefore locate two different ways of controlling the property domain.

A separate minimization concerns de-dicto stability. Full bilateral extension stability

$$STAB_G^\pm : \quad wRz \Rightarrow ext_w G \equiv ext_z G$$

contains more information than the positive theorem uses. I split the positive channel into persistence and reflection:

$$\begin{aligned} Pers_G^+ & : \quad wRz \wedge +ext_w G(x) \Rightarrow +ext_z G(x), \\ Refl_G^+ & : \quad wRz \wedge +ext_z G(x) \Rightarrow +ext_w G(x). \end{aligned}$$

Their conjunction is written $STAB_G^+$. Lean proves that bilateral stability implies $STAB_G^+$, but the two positive directions are exactly what the de-dicto proof consumes: reflection converts de-dicto possibility into frozen de-re possibility, whereas persistence converts frozen de-re necessity back into de-dicto necessity. Consequently the minimized classification-route theorem is

$$\begin{aligned} & G\text{-admissible} + G_F\text{-realization} + A1_{L,F}^{adm} + REG_{G,F}^{neg} \\ & + NE_F\text{-realization} + A5_F^+ + STAB_G^+ + \Diamond^+ \exists^E x G_F(x) \\ & \implies \Box^+ \exists^E x G_F(x). \end{aligned}$$

No negative-extension stability and no S4/S5 frame condition occurs. Replacing the classification block by Ind_G^{adm} , or more structurally by Sat_P^{adm} , gives an alternative theorem with the same positive stability interface.

The positive stability interface can now be derived from a more structural premise. Let $R_{F,adm}^+$ restrict positive-property rigidity to the selected admissible extension domain:

$$Adm(X) \wedge +P_w(X) \wedge wRz \implies +P_z(X).$$

Lean proves the asymmetric pair

$$\begin{aligned} G_F\text{-realization} + R_{F,adm}^+ & \implies Refl_G^+, \\ G_F\text{-realization} + R_{F,adm}^+ + Sym(R) & \implies Pers_G^+. \end{aligned}$$

Forward rigidity is enough for reflection because every extension positive at the source remains positive at the target, where target Godlikeness consumes

it. Persistence requires only the converse positivity transport, not the converse edge itself. Write $R_{F,adm}^{+\leftarrow}$ for

$$Adm(X) \wedge wRz \wedge +P_z(X) \implies +P_w(X).$$

Lean proves

$$\begin{aligned} R_{F,adm}^+ + Sym(R) &\implies R_{F,adm}^{+\leftarrow}, \\ G_F\text{-realization} + R_{F,adm}^{+\leftarrow} &\implies Pers_G^+, \\ G_F\text{-realization} + R_{F,adm}^+ + R_{F,adm}^{+\leftarrow} &\implies STAB_G^+. \end{aligned}$$

Substituting the last implication into the minimized theorem yields positive de-dicto necessity without a frame premise. The older $R_{F,adm}^+ + Sym(R)$ package remains sufficient because it entails the converse transport, but frame symmetry is not part of the sharper proof interface.

The weakening of stability is strict even under a stronger surrounding control theory. A complete two-world S5 fixture satisfies both A1 directions and the original full admissible regularity package. The positive membership set of G is $\{a\}$ at both worlds, but negative information about a non-Godlike individual changes from falsity support to a gap. Hence $STAB_G^+$ holds while $STAB_G^\pm$ fails, and positive necessary de-dicto Godlikeness still holds. This finite witness shows that the positive theorem is genuinely insensitive to drift confined to the negative extension channel.

A second finite regression independently checks the derived bridge over all 256 two-world, one-entity configurations on the negation-closed classical admissible domain $\{\mathbf{T}, \mathbf{F}\}$. Of these, 152 satisfy $R_{F,adm}^+$, and every one satisfies $RefI_G^+$; 152 satisfy $R_{F,adm}^{+\leftarrow}$, and every one satisfies $Pers_G^+$. There are 112 candidates satisfying both transports. Only 80 are symmetric, leaving 32 asymmetric candidates with complete positive G -stability. An explicit one-way frame with non-vacuous positivity witnesses the strict bounded separation from symmetry. A second fixture satisfies converse transport without forward transport, preserving persistence while reflection fails. These are bounded separation results, not global necessity claims.

These results do not establish global minimality. They reveal two distinct recovery architectures inside the same Fitting candidate: one resolves positivity gaps by classification, while the other factors the admissible bilateral property domain through positive-property profiles and thereby makes the gap irrelevant to extensional essence.

8.8 Fitting quotient-domain algebra

The positive-profile saturation condition introduced above has a stronger bilateral consequence. The selected admissible domain is already closed under FDE negation. Hence, if an admissible extension Y is positively constant on a positive-profile equivalence class, then the admissible complement $\neg Y$ is

positively constant on that class as well. Since positive membership in $\neg Y$ is negative membership in Y , Lean proves

$$Sat_P^{adm} \iff Sat_{\pm}^{adm},$$

where $Sat_{\pm}^{adm}$ requires both positive and negative membership of every admissible bilateral extension to be constant on the same positive-profile classes. Thus the quotient interpretation is genuinely bilateral rather than merely a statement about the positive projection of an extension.

The quotient is also stable under the basic FDE extension operations. Define pointwise

$$\begin{aligned} +(X \wedge_F Y)(x) &\iff +X(x) \wedge +Y(x), & -(X \wedge_F Y)(x) &\iff -X(x) \vee -Y(x), \\ +(X \vee_F Y)(x) &\iff +X(x) \vee +Y(x), & -(X \vee_F Y)(x) &\iff -X(x) \wedge -Y(x). \end{aligned}$$

Lean proves that, for each fixed positive-profile quotient, the bilateral extensions that factor through it contain the FDE top and bottom extensions and are closed under FDE negation, conjunction, and disjunction. In this preservation sense they form a De-Morgan-style subalgebra of the full bilateral extension algebra.

This result does not derive profile saturation from ordinary algebraic closure. A finite countermodel makes the distinction explicit. Let the selected domain over two entities be the complete classical four-extension algebra

$$\emptyset, \quad \{a\}, \quad \{b\}, \quad \{a, b\}.$$

It contains top and bottom and is closed under FDE negation, conjunction, and disjunction. Let positivity support only the universal extension $\{a, b\}$. Then a and b have the same positive-property profile, but the admissible singleton $\{a\}$ distinguishes them. Therefore profile saturation fails although the selected extension domain is fully closed under the basic FDE algebra.

Consequently the verified dependency is asymmetric:

$$\boxed{Sat_P^{adm} \implies \text{algebraically stable quotient factorization,}}$$

while

$$\boxed{\text{FDE algebra closure} \not\Rightarrow Sat_P^{adm}.$$

The quotient route therefore still requires an independent extensionality or saturation principle. Algebraic closure is compatible with that principle and preserves it, but does not explain its origin.

This separation also sharpens the comparison with the classical Fitting ultrafilter result [3]. Ultrafilter-style maximality constrains which extensions count as positive; profile saturation instead constrains how the selected extension domain may distinguish entities. The two mechanisms operate on different semantic interfaces, and neither should be silently identified with the other in the four-valued reconstruction.

8.9 Entailment and the profile quotient

The preceding quotient construction raises a natural question: can the profile-saturation condition itself be generated by closing the admissible property domain under Fitting-style extensional entailment? The classical comparison suggests caution. In the mechanized Fitting analysis, the δ -filter is a family of positive extensions: it contains the universal extension, excludes the empty extension, is upward closed under supersets, and is closed under intersections [3]. This is closure of the positivity filter, not closure of the ambient property domain. The present $A2_F^+$ interface already captures the corresponding monotonicity of positive support under extensional entailment.

I therefore distinguish that source-grounded filter behavior from the stronger domain principle

$$\begin{aligned} EntCl^{adm} : \quad & Adm(X) \wedge \forall w \text{Entail}_F^+(w, X, Y) \\ & \implies Adm(Y). \end{aligned}$$

The stronger principle is unsuitable once FDE bottom is admitted. Since $\perp_F$ has no positive members,

$$\text{Entail}_F^+(w, \perp_F, Y)$$

holds vacuously for every world w and every rigid extension Y . Lean consequently proves

$$\boxed{EntCl^{adm} + Adm(\perp_F) \implies \forall Y \text{Adm}(Y).}$$

Thus naive entailment closure of the admissible property universe reconstructs unrestricted bilateral comprehension. Combined with relevant exemplification consistency, the earlier comprehension obstruction returns immediately:

$$\boxed{EntCl^{adm} + Adm(\perp_F) + CONS_{G,F}^{G,adm} \implies \forall w, x \neg G_F^+(x, w).}$$

The distinction between filter closure and domain closure is therefore substantive rather than terminological.

A finite separation strengthens this negative result. Over two entities I take the complete $4^2 = 16$ bilateral extension universe as admissible. Both entities actually exist at one reflexive world, and positive support is assigned exactly to extensions whose positive channel contains both entities. This model satisfies $A2_F^+$, full domain-level entailment closure, closure under the basic FDE extension algebra, and quotient-compatible actual existence. Positivity nevertheless gives the two entities the same positive-property profile while an admissible singleton extension distinguishes them. Hence

$$\boxed{A2_F^+ + EntCl^{adm} + \text{FDE algebra closure} + \text{profile-compatible existence} \not\Rightarrow Sat_P^{adm}}$$

in this explicit finite model. Entailment closure therefore does not generate the missing quotient structure.

The positive result takes a different form. For each world w , define a bilateral profile closure on arbitrary rigid extensions by

$$\begin{aligned} +Sat_w(Y)(x) &\iff \exists y (x \approx_w^+ y \wedge +Y(y)), \\ -Sat_w(Y)(x) &\iff \exists y (x \approx_w^+ y \wedge -Y(y)). \end{aligned}$$

Lean proves that Sat_w is extensive and monotone and is idempotent up to bilateral extensional equivalence. Every $Sat_w(Y)$ respects the positive-profile quotient. More strongly, if Z already respects the quotient and Y is support-included in Z , then

$$Sat_w(Y) \preceq Z.$$

Thus $Sat_w(Y)$ is the least quotient-respecting bilateral extension above Y in the support-inclusion order.

This yields an exact fixed-point characterization of the earlier domain condition:

$$\boxed{Sat_P^{adm} \iff \forall w, Y (Adm(Y) \rightarrow Sat_w(Y) \equiv Y).}$$

The quotient route can therefore be read as selecting admissible properties from the fixed points of a canonical closure operator induced by positive-property observational equivalence.

Actualist quantification introduces one further boundary. Fitting entailment quantifies only over entities that actually exist at accessible worlds. Even if two entities lie in the same positive-profile class, the existence predicate can still distinguish them. I isolate the compatibility condition

$$ExSat_P : \quad wRz \wedge x \approx_w^+ y \implies (E(z, x) \leftrightarrow E(z, y)).$$

Under this premise Lean proves

$$\boxed{Entail_F^+(w, X, Y) \implies Entail_F^+(w, Sat_w(X), Sat_w(Y)).}$$

Hence actualist Fitting entailment descends through the profile closure when actual existence itself factors through the same quotient.

A second finite fixture shows why the existence premise matters. Two entities have the same positive-property profile, but at the sole accessible target world only one actually exists. An extension whose only positive member is the non-actual representative then entails an empty-positive extension vacuously. Profile closure transfers that positive membership to the equivalent actual representative, after which the saturated entailment fails. Thus the existence-compatibility condition is not merely a notational

convenience, although the finite witness does not establish its global necessity for every alternative quotient construction.

The resulting architecture is more precise than the initial entailment-closure hypothesis. Positivity-filter closure remains governed by $A2_F^+$; the admissible property universe should not be closed indiscriminately under entailment; profile saturation is instead a fixed-point condition for a canonical closure operator; and actualist entailment is compatible with that quotient only when actual existence respects the relevant profile classes. This relocates the next open problem from property-domain entailment closure to the principled status of the actualist quotient condition and to the algebraic/filter structure of the closure fixed points.

9 Related Work

The project sits at the intersection of two mature lines of work whose combination motivates the present question. First, Gödel–Scott and related ontological theories have been extensively reconstructed and analyzed with computational tools, including comparisons of Scott, Anderson, and Fitting variants and their behavior with respect to modal collapse [4, 3]. Recent machine-checked analysis further separates positivity rigidity from ultrafilter structure as explanations of collapse [2].

The Anderson/Fitting comparison is especially relevant to Gate 8. The computer-supported analysis of Benz Müller and Fuenmayor distinguishes intensional properties from their extensions and uses this distinction to compare the two non-collapsing variants [3]. In the source IHOML treatment, Scott and Anderson apply positivity to intensional properties, whereas Fitting moves positivity to property extensions and distinguishes de-re from de-dicto formulations. The present project follows that typing discipline rather than modeling Fitting by merely changing axioms over the existing Scott property type.

The four-valued reconstruction introduces an additional issue that is absent from a purely Boolean extension space. If the Fitting property domain is identified with the unrestricted type of all bilateral extensions, that type contains arbitrary glut constructions. In particular, a universal glut extension makes the project’s relevant exemplification-consistency condition incompatible with any positive Godlike witness. The admissible-domain reconstruction in Section 8 should therefore be read as a semantic-domain choice forced by the interaction between Fitting’s extensional architecture and unrestricted bilateral comprehension, not as an attribution to Fitting of an historical four-valued restriction. The selected domain remains closed under FDE negation and may contain genuinely glutty extensions.

This also sharpens the de-re/de-dicto comparison. The project reproduces a K-style, frame-free de-re possibility-to-necessity route at the bilateral

extensional interface, while the de-dicto route requires an explicit stability bridge connecting the extension of the Godlikeness intension across accessible worlds. The source literature already warns against conflating neighboring de-re and de-dicto formulations [3]; the present finite separation model makes that distinction executable in the four-valued setting.

The same literature motivates a methodological distinction in the Anderson branch. Anderson strengthens Godlikeness by classifying the necessary properties of an individual through positivity and modifies essence accordingly, rather than merely adding a stronger Scott-style reflection principle [3]. The bilateral negative clauses used here are consequently a project-specific FDE reconstruction around a literature-grounded positive interface; they are not attributed to Anderson himself.

Second, Belnap–Dunn/FDE semantics provides the canonical four-valued distinction between truth-only, falsity-only, both, and neither [1, 5]. Modal extensions with Belnapian values have been developed using Kripke and algebraic semantics [6, 8]. Particularly close in semantic neighborhood is Petrukhin’s recent treatment of essence and accident modalities in an S5-based four-valued setting [7].

Novelty status. A dedicated prior-art review specifically targeting four-valued or FDE-based reconstructions of the Gödel–Scott positivity axioms, Anderson-style necessary-property definitions, Fitting-style extensional positivity, and restrictions on bilateral property domains remains required before any novelty claim is made. The manuscript therefore states structural and mechanization results only. In particular, the symmetry-only T3 reductions, the bilateral Anderson candidate, the unrestricted Fitting-comprehension obstruction, and the admissible-domain Fitting reconstruction are project results whose relation to all prior non-classical reconstructions still requires the Gate-9 audit.

10 Conclusion and Research Agenda

This manuscript treats the Gödel–Scott theory not merely as an object to be translated from two values to four, but as a controlled test case for locating informational dependencies that classical bivalence suppresses.

By version 0.16 the Scott reconstruction and verification branches reach positive necessary Godlike existence, the associated collapse analysis, and a mechanically supported comparison of three different Godlikeness/essence architectures. Gates 1–4 establish the bilateral propositional and modal semantics, decompose A1 and A4, falsify the original collapse-separation conjecture under involutive FDE negation, and isolate glut/gap failures in the classical reflection step. Gate 5 reconstructs truth-only positive-property exemplification, a control reading of A3, possible positive Godlike existence,

and support-based bilateral Godlikeness.

Gate 6 identifies the sharper Scott bottleneck. A natural bilateral definition of essence recovers Scott D2 classically, but it does not make Godlikeness an essence automatically. Two explicit two-world S5 models satisfy the Gate-5 control conditions together with strong A1 and positive rigidity while refuting

$$T2^+ : \quad +G(x) \Rightarrow +G \text{ Ess } x.$$

The glut and gap models fail for different informational reasons. A targeted regularity package recovers T2, and Gate 7 changes the epistemic status of this result by machine-checking

$$\text{NegExemplification} + G\text{-sup} + A1_L + R^+ + REG_G \Longrightarrow T2^+.$$

Lean also verifies the earlier truth-only possibility theorem

$$A1_R + A2^+ \Longrightarrow T1_T.$$

Thus the division of labor inside Scott's A1 is represented in independent formal derivations.

Once T2 is available, the remaining Scott necessary-existence branch requires less frame structure than the original S5 presentation suggested. Gate 8 proves

$$\begin{aligned} & \text{Sym}(R) + \text{PossibleGod} + T2^+ + A5^+ + NE\text{-sup} + G\text{-sup} \\ & \Longrightarrow T3^+. \end{aligned}$$

Neither reflexivity nor transitivity is a premise. A reflexive-transitive but non-symmetric finite model refutes the corresponding S4 replacement. Reflexivity reappears later, where T3 supplies the global witness condition and where the current essence-compressed positive collapse theorem uses

$$\text{Refl}(R) + T2^+ + T3^+ + G\text{-sup} + CONST \Longrightarrow MC^+.$$

The informative negative collapse schema then follows from the independently machine-checked collapse-channel equivalence.

The finite-model layer provides complementary negative evidence. The broader exhaustive two-world, one-entity G, Z family retains 873 models satisfying the full current Scott T2 recovery package, and every retained model satisfies $T2^+$. Dropping any one of $A1_L$, positive rigidity, relevant positivity completeness, or relevant exemplification consistency yields a T2 countermodel in the same bounded family. The first $A1_L$ -drop witness is now rendered directly from the oracle as a two-node Kripke graph with complete local FDE tables, making both its accessibility relation and its positive/negative support matrix independently inspectable. This remains bounded indispensability, not global necessity.

Gate 8 demonstrates that the T2 bottleneck is definition-sensitive. The project-internal exact positive Godlikeness condition internalizes local positive reflection. Lean consequently proves

$$G\text{-exact}^+ + R^+ \implies T2_{\text{exact}}^+$$

without $A1_L$, $COMP_P^G$, or $CONS_G^G$, while an executable model shows that exact positive Godlikeness remains compatible with genuine gluts.

The Anderson comparison shifts the modal work differently. Its literature-grounded positive interface classifies necessary properties of an individual through positivity and modifies essence accordingly. The project adds bilateral negative-evidence clauses and machine-checks their classical recovery. A complete-S5 model of the candidate has positive necessary Godlike existence while positive modal collapse fails for a contingent formula application. The Anderson T3 branch also reduces to symmetry alone, with a corresponding non-symmetric S4 countermodel. Scott and Anderson therefore share the same currently verified T3 return bridge while differing substantially in the upstream Godlikeness/essence mechanism.

The Fitting comparison exposes a third and more radical axis: the location of the property domain itself. The mechanized literature distinguishes intensions from extensions and applies Fitting positivity to extensions [3]. Enforcing that distinction in the four-valued setting first produced an instructive failure. If the extensional domain is identified with every bilateral support assignment, it contains a universal glut extension. Lean proves that the resulting unrestricted relevant-consistency condition excludes every positive Fitting-Godlike individual. The first unrestricted recovery implication was therefore logically correct but unsuitable as a substantive non-vacuous theorem.

The repaired candidate introduces a selected domain Adm of admissible extensions, closed under FDE negation but not globally constrained to consistency. On this domain Lean proves the non-vacuous Fitting essence theorem

$$\begin{aligned} & G\text{-admissible} + G_F\text{-realization} + A1_{L,F}^{adm} + REG_{G,F}^{adm} \\ & \implies G_F^+(x, w) \Rightarrow Ess_F^+(ext_w G, x, w), \end{aligned}$$

with no positivity-rigidity premise. This is a genuine structural difference from the current Scott-support recovery route: extensional necessary entailment operates on rigid extensions, so positivity need not be transported to each accessible world.

Gate 8 then minimizes this Fitting route further. Full relevant exemplification consistency is stronger than the A1-L proof itself requires: it is sufficient to demand consistency only when positivity completeness classifies the currently possessed extension negatively. Writing this targeted condition

as $NegCons_{G,F}^{adm}$, Lean proves

$$A1_{L,F}^{adm} + COMP_{P,F}^{G,adm} + NegCons_{G,F}^{adm} \implies G_F^+(x, w) \Rightarrow Ess_F^+(ext_w G, x, w).$$

A finite glut model separates the targeted condition from full consistency on the A1-L-only route. The weakening is itself A1-sensitive: restoring $A1_R$ allows Lean to reconstruct full relevant exemplification consistency from positivity completeness plus $NegCons_{G,F}^{adm}$. Thus the two A1 channels again have distinct structural roles even inside the Fitting comparison.

A second Fitting recovery route shows that positivity completeness is itself route-specific rather than globally necessary for the current essence/NE conclusion. The previously isolated Godlike-indiscernibility interface is now derived from a quotient-style condition on Adm . Let

$$x \approx_w^+ y$$

mean that every positively supported admissible extension contains x and y alike. Lean proves this to be an equivalence relation and proves that positive Fitting-Godlike individuals necessarily lie in the same equivalence class. If every admissible rigid extension is saturated with respect to these positive-profile classes, then

$$Sat_P^{adm} \implies Ind_G^{adm},$$

and therefore

$$G\text{-admissible} + G_F\text{-realization} + Sat_P^{adm} \implies G_F^+(x, w) \Rightarrow Ess_F^+(ext_w G, x, w)$$

without A1, $COMP_P^{G,adm}$, or exemplification consistency. A one-world three-entity finite model contains two distinct positive Godlike individuals in one non-singleton profile class, satisfies the saturation condition, and still has a relevant admissible extension with positivity value **N**. Thus the classification gap remains real while the essence/NE route succeeds.

The quotient condition has a more canonical formulation than the initial indiscernibility interface suggested. FDE-negation closure upgrades positive-profile saturation to bilateral factorization, and quotient-respecting bilateral extensions are closed under the basic FDE operations. Ordinary algebra closure does not generate the quotient: a complete classical extension algebra can be closed under negation, conjunction, and disjunction while still containing a singleton that distinguishes two positivity-profile-equivalent individuals.

The entailment experiment sharpens this boundary further. The classical δ -filter closure described in the literature is closure of the family of positive extensions, not closure of the ambient property domain [3]. Naively

imposing global entailment closure on Adm is too strong once FDE bottom is admissible, since bottom positively entails every extension. Lean proves

$$EntCl^{adm} + Adm(\perp_F) \implies \forall Y \text{ } Adm(Y),$$

and, with relevant exemplification consistency, this recreates the unrestricted-comprehension obstruction and excludes every positive Godlike witness. A separate exhaustive finite fixture over all sixteen bilateral extensions on two entities satisfies $A2_F^+$, full domain-level entailment closure, FDE-algebra closure, and profile-compatible actual existence while profile saturation still fails. Thus entailment closure is not the missing source of the quotient.

Instead, Lean defines a canonical bilateral closure Sat_w that sends an arbitrary extension to the union of the positive-profile classes meeting its positive and negative support. This operation is extensive and monotone, idempotent up to bilateral extensional equivalence, and has the universal property of being the least quotient-respecting extension above its input. The earlier saturation principle is exactly the fixed-point condition

$$Sat_P^{adm} \iff \forall w, Y (Adm(Y) \rightarrow Sat_w(Y) \equiv Y).$$

Thus the selected Fitting property domain can be understood as living inside the fixed-point algebra of a canonical observational closure operator.

Actualist entailment introduces one additional interface. If actual existence is constant on positive-profile classes throughout the accessible region, then Lean proves

$$Entail_F^+(w, X, Y) \implies Entail_F^+(w, Sat_w(X), Sat_w(Y)).$$

A two-world finite countermodel shows that this descent can fail when two profile-equivalent individuals are split by the actual-existence predicate. The quotient analysis therefore reaches beyond property membership: actualist domain variation must also be coordinated with the entity quotient if Fitting entailment is to factor through it directly.

The literature's classical Fitting analysis supplies a useful contrast. Fitting positivity is a δ -ultrafilter on rigid extensions [3]. Isolating only its maximality component gives a complement-decision principle. In the bilateral reconstruction, Lean shows that complement decision together with relevant exemplification consistency already reconstructs local positive reflection and hence $COMP_P^{G,adm}$; complement decision together with $A1_R$ also reconstructs $COMP_P^{G,adm}$. The ultrafilter-style route therefore explains why classification is natural in the classical theory, while positive-profile saturation gives a genuinely different four-valued route that leaves relevant positivity gaps unresolved.

The admissible bilateral negative clauses also recover classically. Lean proves complementary recovery for extensional entailment, Godlikeness,

essence, necessary actual exemplification, and necessary existence under classical coherence of the selected extensions. Thus the repair is not merely a positive-fragment construction.

The Fitting necessary-existence branch separates de re from de dicto explicitly. Freezing the current G -extension yields a frame-free theorem:

$$\text{admissible Fitting stack} + \Diamond_{dere}^+ \exists^E x \text{ ext}_w G(x) \implies \Box_{dere}^+ \exists^E x \text{ ext}_w G(x).$$

No reflexivity, symmetry, transitivity, seriality, or R^+ premise is used.

The earlier de-dicto bridge used full bilateral extension stability. The minimized proof reveals that only positive membership in G matters. Let $Pers_G^+$ preserve positive G -membership from the source of an accessibility edge to its target and let $RefI_G^+$ reflect positive membership back from target to source. Their conjunction, $STAB_G^+$, suffices for

$$\begin{aligned} &\text{minimized admissible Fitting stack} + STAB_G^+ + \Diamond^+ \exists^E x G_F(x) \\ &\implies \Box^+ \exists^E x G_F(x), \end{aligned}$$

again without an S4 or S5 frame premise. A complete-S5 finite fixture satisfies both A1 directions and even the older full regularity package while the positive G -extension is stable and the negative G -extension drifts. Positive necessary de-dicto Godlikeness still holds. Hence full bilateral $STAB_G$ is genuinely stronger than the positive theorem needs.

The positive stability assumption itself now has a derived form. Restrict forward positivity rigidity to admissible Fitting extensions as $R_{F,adm}^+$, and write $R_{F,adm}^{+\leftarrow}$ for converse transport of target positivity back to the source along an existing edge. Lean proves that the former derives $RefI_G^+$, while the latter derives $Pers_G^+$, so their conjunction yields all of $STAB_G^+$ without a frame condition. Symmetry plus $R_{F,adm}^+$ entails $R_{F,adm}^{+\leftarrow}$, recovering the older route. A complete 256-case finite regression finds 112 candidates satisfying both transports, including 32 asymmetric candidates. Thus accessibility symmetry is strictly stronger than the bidirectional positivity transport consumed by the proof in this bounded family.

Finally, the admissible Fitting candidate admits the same qualitative anti-collapse phenomenon as the Anderson candidate. A complete two-world S5 model satisfies the encoded Fitting stack and positive necessary actual Godlikeness while a contingent $Q(a)$ refutes MC^+ . The admissible property domain in that model still contains genuine **B** information, so the result is not obtained by silently restoring global bivalence.

The comparative picture is therefore sharper than a hierarchy of stronger or weaker modal logics. Scott, Anderson, and Fitting distribute the decisive

structure differently:

Scott support: reflection recovery and R^+ precede $T2^+$,
 Anderson: necessary exemplification is built into Godlikeness,
 Fitting: positivity/essence range over selected rigid extensions,
 with classification and quotient-fixed-point recovery routes.

The accessibility relation matters, but it is not the only place where modal strength enters the theory.

The main open questions now separate accordingly. For Scott:

Which weaker principled four-valued assumptions, if any,
 suffice for $T2^+$ globally?

For Fitting's quotient semantics:

What independently motivates the fixed-point selection
 and profile-compatible actual existence?

For the ultrafilter comparison:

Can a genuinely four-valued δ -filter/ultrafilter be defined
 on the closure fixed points without restoring *COMP*?

For the remaining de-dicto minimization:

Can $R_{F,adm}^{+\leftarrow}$ be restricted to G-triggered
 or quotient-selected admissible extensions?

Exact higher-order correspondence with the complete published formal developments, paired-neighborhood generalization, and a dedicated prior-art audit remain publication-stage requirements. The paper continues to distinguish general Lean theorems, explicit finite countermodels, bounded exhaustive evidence, and open global-minimality claims.

References

- [1] Nuel D. Belnap. A useful four-valued logic. In J. Michael Dunn and George Epstein, editors, *Modern Uses of Multiple-Valued Logic*, pages 7–37. D. Reidel, 1977.
- [2] Christoph Benzmüller. A comment on modal collapse and ultrafilters in gödel's ontological argument, 2026.

- [3] Christoph Benzmüller and David Fuenmayor. Computer-supported analysis of positive properties, ultrafilters and modal collapse in variants of gödel’s ontological argument. *Bulletin of the Section of Logic*, 49(2):127–148, 2020.
- [4] Christoph Benzmüller and Dana Scott. Notes on gödel’s and scott’s variants of the ontological argument. *Monatshefte für Mathematik*, 208:569–611, 2025.
- [5] J. Michael Dunn. Intuitive semantics for first-degree entailment and coupled trees. *Philosophical Studies*, 29:149–168, 1976.
- [6] Sergei P. Odintsov and Heinrich Wansing. Modal logics with belnapian truth values. *Journal of Applied Non-Classical Logics*, 20:279–301, 2010.
- [7] Yaroslav Petrukhin. Essence and accident modalities meet belnapian truth values. *Studia Logica*, 2026.
- [8] Umberto Rivieccio, Achim Jung, and Ramon Jansana. Four-valued modal logic: Kripke semantics and duality. *Journal of Logic and Computation*, 27:155–199, 2017.